Mistral vs DeepSeek vs LLaMA: Assessing Factuality and Completeness in COVID-19 Medical Summaries

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Abstract

Following the outbreak of COVID-19 in 2020, Coronavirus-related literature became an invaluable source of information for understanding the virus and its associated side effects. At the same time, there has been an increasing reliance on large language models to summarize biomedical text for non-expert users, raising concerns about factual errors or hallucinated claims in important settings. We evaluate the factual reliability of LLM-generated summaries of COVID-19 side effects using a QAGS-style consistency pipeline on 250 scientific papers, comparing LLaMA-3.1-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Qwen-7B under shared prompts and fixed decoding settings. Results show that the Mistral model performs the best compared to the other models. However, the low performance across all models of under a 0.50 mean F1 score indicates poor and inadequate ability to summarize relevant papers.

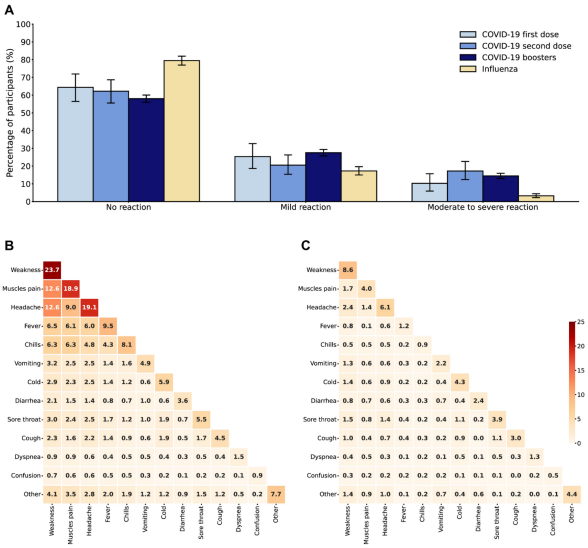


Figure 1: Example of heterogeneous COVID-19 side-effect evidence across outcomes and cohorts. The figure illustrates that adverse-event patterns can vary by severity and population subgroup.

1 Introduction

This project examines how reliably different large language models summarize peer-reviewed scientific papers that report COVID-19 side effects. Our goal is to measure whether these models can produce summaries that are factually accurate and free of hallucinations when working with long and technical biomedical documents from the CORD-19 dataset. Side-effect reporting often includes uncertainty and mixed findings, which makes this a useful setting for testing whether models oversimplify results, omit important details, or introduce information that is not supported by the paper.

A core challenge in side-effect summarization is that biomedical papers typically separate evidence across multiple sections–methods, cohorts, limitations, etc.–so the most important qualifiers might not appear near the headline result. Summaries that sound plausible can still be misleading if they confuse correlation with causation, generalize beyond the studied population, or ignore conflicting findings across studies because of inaccuracies in understanding different sections. This makes COVID-19 side effect papers a particularly stringent test of grounding: producing readable synthesis requires compressing complex evidence while still preserving constraints on interpretation.

In a broader context, both professionals and the general public rely on these strong syntheses to guide important decisions during public health crises. Errors in this setting can often undermine trust in scientific reporting and lead

to misinformation at times when clear communication is most needed. At the same time, COVID-19 side-effect evidence is frequently heterogeneous across studies and evolves over time; Figure 1 illustrates how adverse-event patterns can vary across cohorts and outcome categories rather than supporting a single uniform conclusion. Useful summaries must therefore preserve uncertainty and population-specific qualifiers rather than smoothing findings into overly confident claims. Evaluating models on this task offers a concrete way to assess whether modern LLMs can support responsible scientific communication, producing concise summaries that remain grounded in the original papers while still correctly conveying nuance.

We evaluate three models: LLaMA-3.1-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Qwen-7B. These models differ in parameter scale, training approach,and the level of public documentation available about their development. Comparing the models allows us to see how these differences relate to performance on a challenging medical summarization task. Our first research question is understanding how factually consistent each model's summaries are with the source papers as measured by QAGS-style scoring. Our second question is understanding whether model scale and training choices (i.e. instruction tuning versus distillation) are associated with more reliable summaries and fewer hallucinations. We expect the distilled and instruction-tuned models to differ in their tendency to omit details/introduce unsupported claims, even when prompted under the same decoding settings.

2 Methods

2.1 Datasets

The CORD-19 database, accessible via Kaggle, is used to determine the efficacy of LLMs on summarizing medical information on COVID-19 side effects. CORD-19, otherwise known as the COVID-19 Open Research Dataset, was assembled and released as a large data source of coronavirus research papers by the Allen Institute for AI in collaboration with various organizations invested in computer science and medicine, as well as the White House (Allen Institute For AI, 2020). According to the Kaggle site, the dataset was initially started on March 13, 2020, when Covid-19 became more prevalent in the United States. The final update was added on June 2, 2022. The papers included within the dataset were collected using Semantic Scholar, all sourced from various reliable publishers of research papers, such as PubMed Central (Wang LL, 2020). The publishers released the dataset via Kaggle as a competition to involve the general public to answer vital Covid-related questions via Natural Language Processing.

The data is accessed by downloading the dataset via kagglehub in Google Colab. Because the data is saved as JSON objects containing information about the papers, including document text and author names, we iterated through the JSON object and saved relevant elements, namely the paper ID and paper text, in a DataFrame to make the data more accessible for later tasks (see Table 1 and Table 2). To further ensure that only relevant data is used within our evaluation regarding the summarization side effects associated with COVID-19, we explicitly filter for any documents that contain the terms "side effects", "adverse effects", "complications" or "symptoms."

Before data selection, there are about 717,000 JSON files containing data from unique documents discussing coronavirus, with specifically 401,270 files available as PDF and 315,742 files available as PMC. Due to the resource-demanding size of the original corpus, we limit the size of the database that we parse through, as we ran into RAM space issues that would crash the session once more than about 100,000 papers were processed by our code. Thus, we parse through 100,000 PMC papers and add papers that mention terms related to "side effects" to our final dataset to be used for evaluation. With this approach, we have a final dataset with 51,020 unique research papers that contain terms related to "side effects" in their discussion of COVID-19.

We plan on using the full text of relevant

Dataset	Size	Input length
CORD-19 (reduced)	250 papers	8000 tokens

Table 1: Dataset statistics for the curated CORD-19 subset used in our experiments.

documents as the input to the LLMs, as our goal is to generate summaries from these documents. Within the subset of papers that mention terms related to "side effects" of coronavirus, the shortest paper text has a length of 133 characters, whereas the longest paper text has a length of 925,965 characters. The mean length of the texts is about 24,796 characters, and the median length of the texts is about 21,365 characters.

Due to memory restrictions on Colab, the final dataset was cut down to 250 research papers. This subset of research papers has a mean length of about 25,603 characters and a median length of about 21,889 characters. The shortest paper within this subset is 2348 characters long, and the longest paper is 93,031 characters.

2.2 Models/Approach

We originally planned to evaluate a mix of open-weight and closed-weight systems, including GPT-OSS-20B and Gemini 1.5 Flash. In practice, Gemini was constrained by API quota and rate limits while GPT-OSS-20B failed to run reliably in our environment due to memory and runtime issues during model loading. To ensure reproducibility and enable evaluation on our sample of documents, we restrict our final experiments to locally runnable, instruction-tuned models that can be executed more consistently under fixed decoding settings.

Our final model set consists of LLaMA-3.1-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Qwen-7B. Although all three models are open-weight or research-accessible, they originate from distinct development ecosystems and differ in training approach, scale, and optimization priorities. This allows us to test whether factual consistency and hallucination patterns are stable across model families rather than specific to a single developer or training pipeline.

Paper ID	Paper Text
PMC7787702	As it evolves, the COVID-19 pandemic continues to impact nearly every aspect of clinical practice. With initial restrictions lifted and thousands of recovered COVID patients reentering the community, many of these patbuildingients are presenting for elective and non-urgent surgeries. However, there is limited evidence regarding the possible impact the disease may have on surgical risk and recovery. COVID-19 has revealed itself to cause multi-organ system disease, at times severe, with a protean list of clinical manifestations. Effects are not limited to systemic disease—prolonged hospital stays and post-viral syndromes are sequelae that should be taken into account when evaluating a patient for elective surgery. Previous studies have illustrated the potential for high morbidity and mortality in patients who undergo procedures while COVID positive, ...
PMC8294891	Rheumatoid arthritis (RA) is the most common autoimmune inflammatory arthritis.[1] RA can lead to progressive joint disability, systemic inflammation, anemia and cardiovascular disease. The prevalence of RA ranges from 0.4% to 1.3% and is associated with gender (2–3 times higher in women than men) and region of population residence (higher in the north than in the south and higher in urban people than in rural areas).[2–6] Currently, there are more biological disease-modifying anti-rheumatic drugs (DMARDs), ...

Table 2: Example data from final dataset after filtering CORD-19 dataset on terms related to "side effects."

2.2.1 LLaMA-3.1-8B-Instruct (Meta, open-weight)

First, we use LLaMA-3.1-8B-Instruct, an 8-billion parameter decoder-only transformer model released by Meta as part of the LLaMA 3.1 family [1]. The model is pretrained as a general purpose language model and instruction-tuned to improve prompt following and downstream task performance. Meta provides a high-level description of the training process, indicating the model is trained on a mixture of publicly available, licensed, and synthetic

[1] https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct

data. However, they withhold detailed information about dataset composition and weighting. The model is accessed via the Hugging Face model hub and run locally using officially released pretrained weights with bfloat16 precision, without additional fine-tuning.

2.2.2 DeepSeek-R1-Distill-Qwen-7B (DeepSeek, open-weight)

Our second model is DeepSeek-R1-Distill-Qwen-7B, a 7-billion-parameter decoder-only transformer distilled from a larger reinforcement-learning optimized teacher model in the DeepSeek-R1 family[2]. The distillation process involved training the smaller model to imitate the teacher's responses on a broad collection of instruction data and generated reasoning traces, producing a compact system that retains much of the teacher model's behavior while remaining computationally efficient (Liu et al., 2025).

This model represents a distinct training paradigm relative to standard instruction tuning and allows us to assess how distillation may affect summarization. We access and run the model locally using its open-weight release.

2.2.3 Mistral-7B-Instruct (Mistral AI, open-weight)

The third model evaluated is Mistral-7B-Instruct, a 7-billion parameter decoder-only transformer released by Mistral AI under the Apache 2.0 license [3]. The model is pretrained on large-scale text data and subsequently instruction-tuned to follow natural language prompts. Mistral emphasizes architectural and training optimizations that enable strong performance at small model sizes (Jiang et al., 2023).

We include Mistral-7B-Instruct as a stable and computationally efficient alternative after encountering runtime limitations with larger open-weight models. Its independent development pipeline provides an additional comparison point for understanding how architectural choices affect factual accuracy in summaries.

[2] https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Qwen-7B

[3] https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3

2.3 Experiments

Our experiments are designed to evaluate how reliably large language models summarize COVID-19 related side effects reported in peer-reviewed scientific literature. The central research question is whether different large language models vary in their ability to produce summaries that are factually accurate, medically complete, and faithful to uncertainty expressed in scientific sources. To address this question, we designed an experimental workflow consisting of dataset construction, summary generation utilizing multiple models, automated evaluation and comparative analysis across models.

The dataset is constructed using the CORD-19 corpus. We load the CORD-19 JSON files through KaggleHub into a Pandas DataFrame and extract paper identifiers and full paper text. To prevent system memory exhaustion within Google Collab, we restricted parsing to 100,000 PMC entries. Preliminary testing indicated that parsing the full corpus exceeded available RAM. From this subset, we filtered papers that are likely to discuss COVID-19 side effects by retaining documents, whose body text contained key words such as "side effects", adverse effects", "complications", or "symptoms". We additionally apply a content–length threshold to remove documents that are too short to support meaningful summarization. This filtering process yielded an accurate dataset focused on scientific descriptions of COVID-19 related side effects.

The dataset size evolves iteratively over the course of this project. We begin with an initial set of 15 papers to validate the preprocessing pipeline, prompt, formatting, and summary generation code. We then expand the dataset to 100 papers. The final experiments are conducted on a dataset of 250 papers, this size was chosen based on time and computational limitations.

We generated summaries using models LLaMA-3.1-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Qwen-7B. All models are prompted using the same base instruc-

tion. The prompt asks the model to summarize each research paper with a focus on the side effects or adverse outcomes associated with COVID-19 and to reflect what is known, uncertain, or disputed in the paper. Based on instructor feedback, we planned to revise the prompt to explicitly include suggested or tentative findings, so the models would be encouraged to capture weakly supported evidence. The revised prompts would have asked models to reflect what is known, suggested, uncertain or disputed in the paper. However, due to time constraints, this prompt modification was not fully incorporated into the final study.

Summary generation for local models is implemented using the Hugging Face Transformers library. For each paper, we constructed a prompt by concatenating the instruction and paper text. Inputs were truncated to a maximum of 8,000 tokens to remain within model context limits. To reduce variability in outputs, we use deterministic decoding. The temperature was set to 0.2, and the maximum number of newly generated tokens was set to 1,024. Models were loaded with automatic device mapping and bfloat16 precision to reduce GPU memory usage. All model versions, prompts and generated outputs are logged to ensure reproducibility.

We measure factual consistency using an implementation inspired by the QAGS framework (Wang et al, 2020). For each summary, we first generated a fixed number of short, factual questions conditioned on the summary text. Question generation was performed using a single shared question generator–FLAN-T5-small–to ensure that all models were evaluated using an identical set of questions and to avoid model-specific biases in question formulation. Each generated question is then answered twice using a pre-trained question answering model, once with original paper text as context and once with the summary text as context. We compared the corresponding answers using an answer-level semantic similarity metric (BERTScore F1), and computed the QAGS score as the average similarity across all questions for a given summary. Higher QAGS scores indicate stronger factual alignment between the summary and source text.

To make evaluation possible on long CORD-19 papers and to keep the protocol consistent across models, we used a fixed question budget of k=3 questions per summary and applied length limits to both the source document and the summary by truncating inputs to a maximum number of characters before question answering and question generation. Specifically, we truncated the paper text to a maximum of 16,000 characters and truncated each model-generated summary to 4,500 characters. This modified setting follows the same QAGS logic but adapts it to long-context biomedical articles, where full-text inputs may exceed the context window of the QA and question-generation models.

In some cases, QAGS scores were undefined (NaN) because the evaluation pipeline produced no valid question–answer pairs for a given summary. This occurred when the question generator failed to produce usable questions or when the QA model returned empty answers for all generated questions on either the truncated source text or the truncated summary. Thus, we report mean QAGS both excluding undefined values and treating undefined values as zero to assess how sensitive our conclusions are to these evaluation failures.

3 Results/Analyses

Figure 1 and Table 3 display the performances of LLaMA-3.1-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Qwen-7 evaluated using the QAGS framework after generating summaries of Covid-19 side effects. Two results were generated per model as F1 Scores: the mean score for all rows (henceforth referred to as "overall score"); and the mean score for rows that do not appear as "NaN" (henceforth referred to as "overall score, excluding NaNs"). Mistral-7B-Instruct has the best performance for both the overall score at 0.2138 and the overall score, excluding NaNs, at 0.4529. DeepSeek-R1-Distill-Qwen-7 has an overall score of 0.1246 and an overall score, excluding NaNs, of 0.3313. LLaMA-3.1-8B-Instruct has an overall score of 0.1126 and an

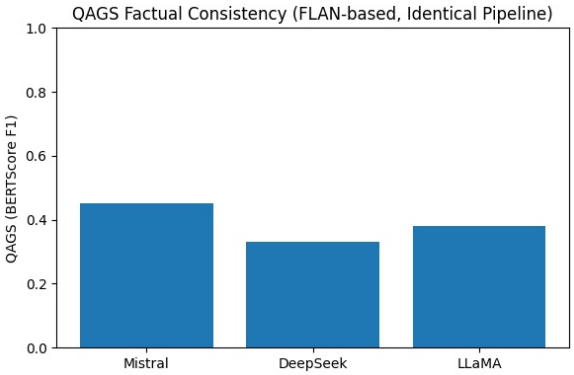


Figure 2: Comparison of models F1 Score performance measurement mean, calculated following the QAGS framework

overall score, excluding NaNs, of 0.3805.

Across all models, excluding the NaNs resulted in a higher score. However, removing NaN rows significantly reduced the number of summaries being evaluated, with all models having more than half of their total summaries discarded when NaNs are not included in the scoring. None of the models performed at or near a level of accuracy and consistency that was expected out of these models, considering the frequent usage and reliance on these models by users to generate accurate summaries. When evaluated with the QAGS framework using the F1 score on datasets with the NaN values included and excluded, none of the models in either approach performed above a 50% F1 performance.

The results show a clear pattern in how the models handle scientific uncertainty. All three models often changed cautious or tentative language in the papers into confident statements turning phrases like "these findings suggest" into "these findings show." This shift makes the summaries sound more certain than the original research actually was. It suggests that the models prioritize sounding decisive instead of accurately reflecting the nuance in the text. In medical research, this is a serious issue because it can make early or inconclusive findings seem definitive. These patterns show that improving factual accuracy in scientific summarization is necessary to represent uncertainty and avoid overstating results.

Model	Total	Scored	$QAGS_0$	QAGS
Mistral	250	118	0.2138	0.4529
DeepSeek	250	94	0.1246	0.3313
LLaMA	250	74	0.1126	0.3805

Table 3: QAGS evaluation results. **Scored** = total rows excluding NaNs. $QAGS_0$ treats undefined (NaN) scores as 0; QAGS excludes NaNs.

4 Related Work

In conjunction with the CORD-19 dataset, NLP has provided researchers a new approach to analyzing COVID-19 through text-based experiments. Tan et al. utilized CORD-19 to fine-tune models to generate extractive summaries (using clustering) and more difficult abstractive summaries, including a multi-step text generation process that trains models using the abstract of a paper and keywords from the full text (Tan et al., 2020). Notably, Tan et al. used an early version of CORD-19 with around 35,000 full-text documents, which likely differs from the version we use. They also did not explicitly filter for terms such as "side effects," used different models (DistilBERT and DistilGPT2), and a different evaluation metric (ROUGE). However, similar to what was observed by Tan et al. we anticipate hallucinated outputs in our own experiment.

For evaluation, we utilize QAGS, which provides a framework to quantify the quality of summaries generated by our models (Wang et al., 2020). Following Wang et al., we input the full text of research papers related to COVID-19 side effects, prompt models to summarize side effects, generate questions from each summary, and then generate two sets of answers using both the original paper and the generated summary. Comparing these answers allows us to measure how well the summary captures vital information and to assess factuality and hallucinations.

Beyond this prior work, researchers have examined factual consistency issues in biomedical summarization more generally. Mullick et al. show that persona-specific prompts (doctor-, patient-, or general-audience) strongly influ-

ence the level of detail and type of information included in a summary (Mullick et al., 2024); accordingly, we used a neutral, non-persona prompt to keep the task focused on content accuracy. Relatedly, Sapkota et al. evaluate transparency and accessibility across leading LLMs and show substantial variation in how models disclose training data and fine-tuning processes, which can affect reproducibility and truthfulness (Sapkota et al., 2025). While our final model set is locally runnable, this broader point still motivates comparing models from different development pipelines.

Long-context approaches have also been applied to COVID-19 scientific literature. Chen et al. (2022) demonstrate that expanding context windows can reduce omission errors when summarizing CORD-19 papers, but hallucination rates remain nontrivial, motivating our comparison across model families (Chen et al., 2025). Karotia and Susan introduce CovSumm, an unsupervised hybrid summarization model for CORD-19 that performs competitively under ROUGE, but it focuses on extractive sentence ranking and does not test factual grounding or detect hallucinated claims (Karotia and Susan, 2023). Finally, Fernández-Pichel et al. find that search engines and large language models often produce confident but unverifiable statements for health-related questions, especially around symptoms or complications (Fernández-Pichel et al., 2025). Although their setting is short-form question answering rather than long-document summarization, their results support the broader concern that models may introduce unsupported claims in medical contexts, which our project evaluates directly by checking summaries against the source papers.

5 Conclusion

This project examined how reliably large language models summarize peer-reviewed scientific papers describing COVID-19 side effects, with a focus on factual consistency and hallucination. Using a QAGS-style evaluation pipeline, we compared LLaMA-3.1-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Qwen-7B under shared prompts and fixed decoding settings to assess how differences in model scale and training approach relate to the faithfulness of summaries. Our results show that Mistral performs the best compared to the DeepSeek and LLaMa models. However, F1 scores generated from all the models fall below 0.5. When taking into account the narrow, critical context of using these models for health-related information, models' performances should be nearly perfect. These results indicate that LLaMA-3.1-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Qwen-7 are not yet at the performance needed to reliably and safely handle the summarization of literature on Covid-19 side effects. The reliance on these models to provide medical information on the side effects of Covid-19 when provided a relevant paper appears to result in a lack of correct and/or vital information in an uninformed response returned to users.

Despite these insights, our study has several important limitations. Compute and memory constraints in Colab, as well as the time required to generate and evaluate long-document summaries, reduced the scope of our experiments. Specifically, we only evaluated 250 papers, which may limit how well our findings generalize to the broader CORD-10 corpus. We also originally planned to use Gemini 1.5 Flash as a closed-weight baseline and GPT-OSS-20B as a larger open-weight model, but Gemini was constrained by API quota and rate limits and GPT-OSS-20B could not be loaded reliably because of GPU memory and runtime issues in our environment.

Future work should use a larger subset of CORD-19 papers to evaluate the summarization abilities of the three tested LLMs. Additionally, experiments that vary prompt structure or section-level summarization (i.e. only summarizing the Results or Discussion of papers) could help isolate which modeling choices drive hallucinations versus omissions. With additional compute and more flexible API access, we could also revisit Gemini 1.4 Flash and GPT-OSS-20B, enabling direct comparisons between API-only commercial systems and locally runnable open-weight baselines at

different parameter scales for COVID-19 side effect summarization.

Overall, this project highlights the balance between accessibility and accuracy in how LLMs communicate medicine. LLMs can make research more understandable for wider audiences but they can also risk oversimplifying complex findings. As these models become more common tools for information sharing, it is important to think about how to build them responsibly. This means that evaluation methods and being careful about how AI-generated summaries are used in public or clinical settings. Our findings serve as a reminder that reliable scientific communications depend on both fluency and truthfulness.

6 Team Contribution Statement

Diya wrote the introduction, research questions, hypothesis, part of the results/analyses section, part of the conclusion, and worked on the team contribution statement. Diya also worked on generating summaries for LLama and determined the LLama model alternative to the Gemini model initially planned.

Elena worked on the datasets section by downloading and filtering CORD-19 JSON files, generating the basic statistics, and selecting example samples; helped write the Abstract section; added code to test LLaMA (following Sachi's template) and experimented with running LLaMA and generating initial summaries; and wrote the Results/Analyses section and the results portion and some limitations of the Conclusion.

Sachi worked on the Models and Approach section, researching each of the models and determining an alternative options when we hit runtime issues; created a template in Colab to run the models and implemented the QAGS-style evaluation pipeline; helped clean model output, analyze NaN cases, and produce results plot; contributed to writing the Conclusion.

Adneen worked on generating summaries for DeepSeek and Mistral, experimented with analysis for DeepSeek (following Sachi's template), and wrote the Experiments Section. All members contributed equally to the related work section by reviewing two papers each.

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